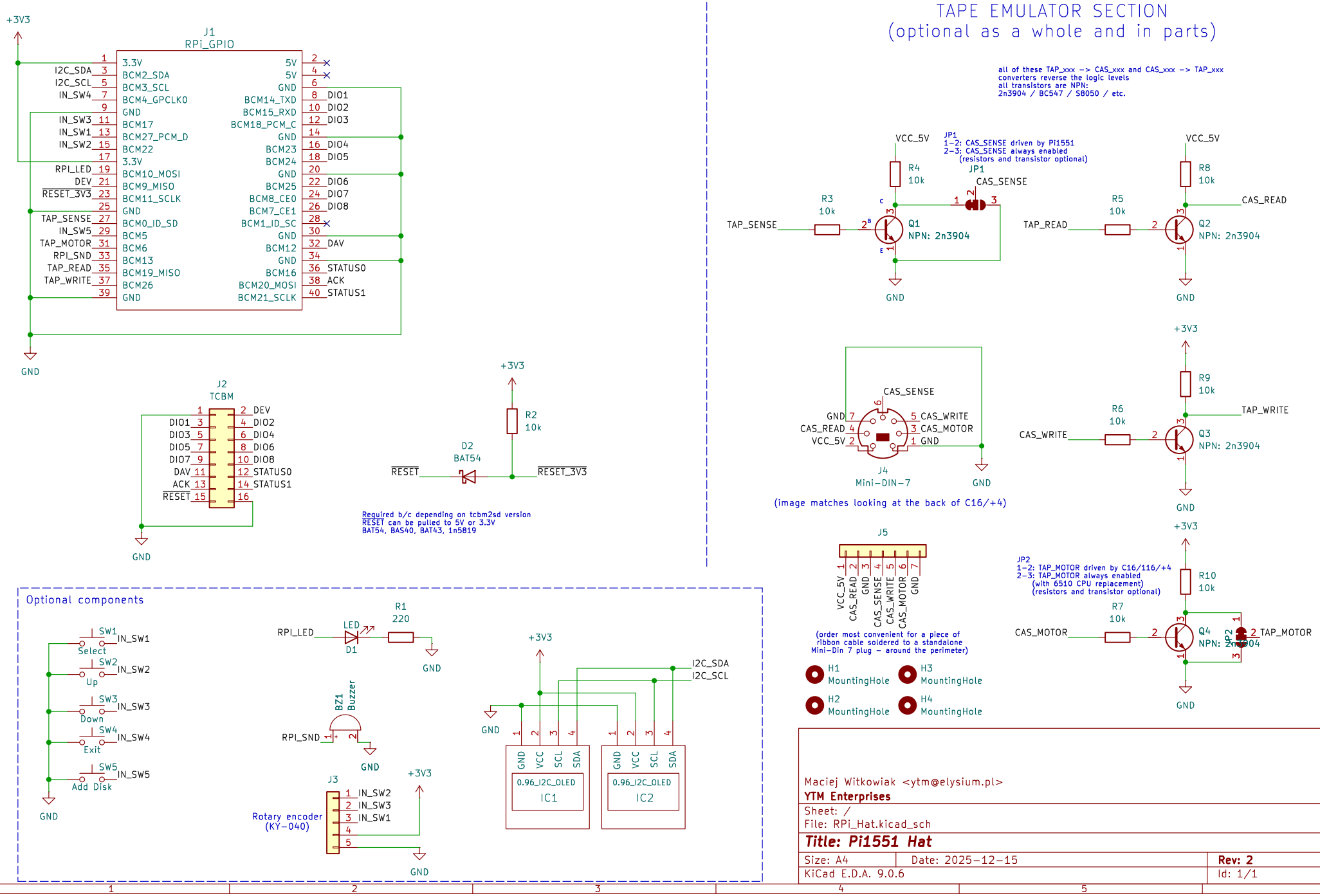
TAPE EMULATOR SECTION
(optional as a whole and in parts)
all of these TAP_xxx -> CAS_xxx and CAS_xxx -> TAP_xxx converters reverse the logic levels
all transistors are NPN:
2n3904 / BC547 / S8050 / etc.
J1
RPi_GPIO
+3V3
GND
I2C_SDA
I2C_SCL
IN_SW4
IN_SW3
IN_SW1
IN_SW2
RPI_LED
DEV
RESET_3V3
TAP_SENSE
IN_SW5
TAP_MOTOR
RPI_SND
TAP_READ
TAP_WRITE
DIO1
DIO2
DIO3
DIO4
DIO5
DIO6
DIO7
DIO8
DAV
ACK
STATUS0
STATUS1
J2
TCBM
R2
10k
D2
BAT54
RESET
RESET_3V3
Required b/c depending on tcbm2sd version
RESET can be pulled to 5V or 3.3V
BAT54, BAS40, BAT43, 1n5819
VCC_5V
JP1
1-2: CAS_SENSE driven by PI1551
2-3: CAS_SENSE always enabled
(resistors and transistor optional)
R4
10k
R3
10k
Q1
NPN: 2n3904
R8
10k
R5
10k
CAS_READ
Q2
NPN: 2n3904
J4
Mini-DIN-7
CAS_SENSE
CAS_WRITE
CAS_MOTOR
(image matches looking at the back of C16/+4)
R9
10k
R6
10k
Q3
NPN: 2n3904
J5
(order most convenient for a piece of ribbon cable soldered to a standalone Mini-Din 7 plug - around the perimeter)
JP2
1-2: TAP_MOTOR driven by C16/116/+4
2-3: TAP_MOTOR always enabled
(with 6510 CPU replacement)
(resistors and transistor optional)
R10
10k
R7
10k
Q4
NPN: 2n3904
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
Optional components
SW1
Select
SW2
Up
SW3
Down
SW4
Exit
SW5
Add Disk
R1
220
LED
D1
BZ1
Buzzer
J3
Rotary encoder
(KY-040)
0.96_I2C_OLED
IC1
IC2
GND
VCC
SCL
SDA
Maciej Witkowiak <ytm@elysium.pl>
YTM Enterprises
Sheet: /
File: RPi_Hat.kicad_sch
Title: Pi1551 Hat
Size: A4
Date: 2025-12-15
Rev: 2
KiCad E.D.A. 9.0.6
Id: 1/1